

Tracking Accuracy and Robustness of a Planar Two-DOF Robot Arm

A Reproducible Comparison of PID, Gravity Compensation, and Computed Torque Control

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Abstract

This study compares independent-joint PID, PID with gravity compensation, and computed torque control (CTC) on a robot moving in a vertical plane under a shared trajectory, sampling interval, and torque limit. Gains are selected on a separate training trajectory and frozen before 42 deterministic evaluation runs covering nominal conditions, unknown endpoint payloads, controller-model parameter errors, and external endpoint force. Nominal joint RMSE values are 1.115037 / 0.152634 / 0.004992 degrees, and formal disturbance recovery delays are 1.555 / 0.087 / 0.263 seconds, in the same controller order. CTC achieves the smallest nominal tracking error, but its advantage does not generalize to all payload, model mismatch, or disturbance cases. Saturation and degraded runs are retained. Control effort, absolute mechanical work, and baseline-relative disturbance diagnostics are explicitly distinguished. Tables, numerical statements, and figures are assembled automatically from accepted results, without retuning the controllers.

Keywords: two-DOF robot arm; computed torque control; gravity compensation; model mismatch; reproducible simulation.

1 Introduction

Robot models reduce the dynamic errors that feedback must compensate, but introduce dependence on model accuracy. This project compares three controllers on the same arm and trajectory under nominal conditions, unknown payloads, controller-model errors, and external force.

The charter predicted that CTC would track best nominally, degrade fastest with unknown payload, and lose its advantage over PID as model error grows; gravity-compensated PID would recover fastest after force removal. Predictions are tested, not used to select results. Robustness here describes finite predefined scenarios, not a universal stability proof.

The theoretical basis is Modern Robotics torque control. The project's implementation defines its particular PD-type CTC, PID integral update, sampling, and saturation; the teaching material does not establish these experimental results.

2 Model

2.1 Coordinates and Kinematics

The arm moves in a vertical plane. Positive x points horizontally right and positive y vertically upward. The angle q_1 is measured counterclockwise from the horizontal axis; q_2 is the angle of the second link relative to the first. Internal calculations use SI units and radians; plots may display degrees.

$$x = l_1 \cos q_1 + l_2 \cos(q_1 + q_2), \quad y = l_1 \sin q_1 + l_2 \sin(q_1 + q_2).$$

End-effector velocity satisfies $\dot{p} = J(q)\dot{q}$. A force in the base frame produces joint torque $\tau_{\text{ext}} = (J(q)^T)F$. The force mapping uses the actual configuration, not the reference configuration.

2.2 Rigid-Body Dynamics

$$M(q)\ddot{q} + c(q, \dot{q}) + G(q) = \tau + \tau_{\text{ext}}.$$

Let $q_{12} = q_1 + q_2$. Without payload, the independent mass-matrix entries are

$$\begin{aligned} M_{11} &= I_1 + I_2 + m_1 l_{c1}^2 + m_2(l_1^2 + l_{c2}^2 + 2l_1 l_{c2} \cos q_2), \\ M_{12} &= M_{21} = I_2 + m_2(l_{c2}^2 + l_1 l_{c2} \cos q_2), \\ M_{22} &= I_2 + m_2 l_{c2}^2. \end{aligned}$$

With $h = m_2 l_1 l_{c2} \sin q_2$,

$$c = \begin{bmatrix} -h(2\dot{q}_1\dot{q}_2 + \dot{q}_2^2) \\ h\dot{q}_1^2 \end{bmatrix}, \quad G = g \begin{bmatrix} (m_1 l_{c1} + m_2 l_1) \cos q_1 + m_2 l_{c2} \cos q_{12} \\ m_2 l_{c2} \cos q_{12} \end{bmatrix}.$$

The endpoint payload is a point mass m_p . It adds $\Delta M = m_p(J^T)J$ and corresponding Coriolis/centrifugal and gravity terms. Specifically, h gains $m_p l_1 l_2 \sin q_2$, and G gains $m_p g[l_1 \cos q_1 + l_2 \cos q_{12}, l_2 \cos q_{12}]^T$. Payload therefore changes inertia, motion coupling, and gravity, rather than merely adding constant torque.

Table 1: Nominal parameters read from the accepted nominal YAML.

Parameter	Value	Unit
l1	0.5	m
l2	0.4	m
lc1	0.25	m
lc2	0.2	m
m1	2	kg
m2	1.5	kg
I1	0.041666667	kg·m ²
I2	0.02	kg·m ²
g	9.81	m/s ²

3 Trajectory

Each segment uses rest-to-rest quintic time scaling. With normalized time $u = (t - t_0)/T$,

$$s(u) = 10u^3 - 15u^4 + 6u^5, \quad q_d = q_a + (q_b - q_a)s(u).$$

Velocity and acceleration follow from analytic derivatives and are zero at every waypoint. The training trajectory is $[-35^\circ, -45^\circ] \rightarrow [50^\circ, 60^\circ]$ over 4.0 s, used only for gain selection. The held-out test trajectory is listed below. All four experiment families use identical test trajectories and initial states.

Table 2: Held-out test waypoints.

Time (s)	q_1 ($^\circ$)	q_2 ($^\circ$)
0.0	20	-35
2.0	-40	25
4.0	55	45
6.0	5	-50
8.0	-30	15

The initial position equals the first waypoint; initial velocity and PID integral error are zero. Error statistics include the initial transient, with no warm-up exclusion or removal of unfavorable intervals.

4 Controllers

Let $e = q_d - q$ and $\dot{e} = \dot{q}_d - \dot{q}$. All gain matrices are diagonal.

Independent-joint PID: $\tau_{\text{req}} = K_p e + K_d \dot{e} + K_i z$. Conditional integration freezes a joint's integral when that joint is saturated and $\tau_i e_i > 0$; otherwise, it advances the integral using the current error. This prevents accumulation that would worsen saturation.

PID with gravity compensation: $\tau_{\text{req}} = K_p e + K_d \dot{e} + K_i z + \hat{G}(q)$. Compensation uses the controller's internal model. Plant and controller parameters are stored separately, allowing independent model errors.

CTC:

$$\tau_{\text{req}} = \hat{M}(q)(\ddot{q}_d + K_d \dot{e} + K_p e) + \hat{c}(q, \dot{q}) + \hat{G}(q).$$

This CTC has no integral term. With an exact model, no saturation, and ideal continuous control, the error satisfies

$$\ddot{e} + K_d \dot{e} + K_p e = -M(q)^{-1} \tau_{\text{ext}}.$$

Unknown external force still drives error: nominal model compensation does not compensate unknown force. The experiments use sampled control, so this ideal continuous equation is not an exact identity of the discrete simulation.

All requested torques pass through a common actuator limit of ± 20.0 N·m per joint. PID gains K_p, K_d, K_i have units N·m/rad, N·m·s/rad, and N·m/(rad·s), respectively. CTC gains K_p, K_d have units s^{-2} and s^{-1} , so numerical gain magnitudes alone cannot compare control strength.

Table 3: Frozen gains, with entries ordered by joint. Display rounding does not change simulation parameters.

Controller	Kp	Ki	Kd
PID	(270.65, 102.72)	(95.432, 64.325)	(16.835, 33.577)
PID+G	(288.13, 193.82)	(16.906, 71.103)	(43.649, 27.903)
CTC	(106.34, 142.6)	N/A	(24.42, 22.309)

5 Methodology

5.1 Tuning and Fairness

Each controller uses 300 candidates, totaling 900, with seed 20260905. PID variants search joint gains directly; CTC is parameterized by natural frequency and damping ratio. All share the training trajectory, initial state, time step, torque limits, and objective:

$$S = 0.7E_n + 0.2U_n + 0.1r_{\text{sat}}.$$

E_n is the overall RMSE after each joint's error is normalized by its training-trajectory angular span; $U_n = \int \sum_i \tau_i^2 dt / (T \sum_i \tau_{\text{max},i}^2)$. Failed candidates are excluded; ties are resolved by control effort and then candidate ID. Screening rules and all candidates are retained in the frozen tuning file. This controls search budget, but parameterization and search ranges may still affect rankings; global optimality is not established.

5.2 Scenarios and Integration

Table 4: Fixed scenario grid. Model errors change only controller parameters $\hat{m}_2, \hat{I}_2$; the plant remains nominal.

Scenario	Condition	Runs
Nominal	Identical plant and controller models	3
Payload	0.0, 0.25, 0.5, 0.75, 1.0 kg; payload added to the plant only	15
Model uncertainty	-30%, -20%, -10%, 0%, 10%, 20%, 30%	21
Disturbance	$F=[10.0, 0.0]$ N, $[4.5, 4.7]$ s	3

The fixed RK4 step and control period are both 0.001 s. Motor torque remains constant within each control interval. In disturbance experiments, constant base-frame force is mapped through $(J(q)^T)F$ at the actual q of every RK4 substep. Each run contains 8001 state samples; the endpoint is recorded without further integration. Control effort includes motor torque only, not external torque.

This reporting stage reads accepted histories without repeating simulations. Existing automated tests cover the model, simulator, controllers, and Stage 21 analysis. They verify implementation and do not replace independent physical experiments.

5.3 Metrics and Sources

$$E_q = \sqrt{\frac{1}{2N} \sum_{k=1}^N \sum_{i=1}^2 e_{k,i}^2}, \quad E_p = \sqrt{\frac{1}{N} \sum_{k=1}^N \|p_d - p\|_2^2}.$$

Peak end-effector error is the maximum Euclidean distance in the specified window. Control effort $U = \int \sum_i \tau_i^2 dt$, in $(\text{N}\cdot\text{m})^2\cdot\text{s}$, **is not energy**. Absolute total mechanical work $W = \int |\sum_i \tau_i \dot{q}_i| dt$, in J, is not electrical consumption either. Integral metrics use actual applied torque and interval-start samples. Saturation fraction measures the duration of intervals with either joint saturated.

Formal recovery is measured from force removal: error must remain at or below 10.0 mm for at least 0.5 s. The reported delay is the start of the first qualifying window. Recovery not confirmed within the record remains unconfirmed and is not reported as zero seconds. Nominal-relative deviations in the figures are diagnostic and do not replace this metric.

Stage 21's `summary.csv` and figures supply the report's numerical results. Their manifest hashes are checked before assembly; experiment parameters come from the same source YAML files. All scenarios are retained. These deterministic runs provide no independent random replicates, so no statistical error bars, confidence intervals, or significance tests are constructed.

6 Results

6.1 Nominal Tracking

Table 5: Nominal accuracy and control input. RMSE aggregates both joints; work integrates absolute total power.

Controller	RMSE (°)	EE RMSE (mm)	Effort	Work (J)
PID	1.115037	25.3661	1440.970	54.452
PID+G	0.152634	3.5193	1444.343	53.683
CTC	0.004992	0.0372	1446.349	53.497

The positions in Figure 1 closely follow the reference, while Figure 2 reveals distinct tracking errors. Nominal RMSE values are 1.115037 / 0.152634 / 0.004992 degrees for PID, PID with gravity compensation, and CTC, respectively. CTC has the smallest error and gravity-compensated PID is intermediate. Control efforts remain similar, so improved accuracy should not be described as a proportional reduction in energy consumption.

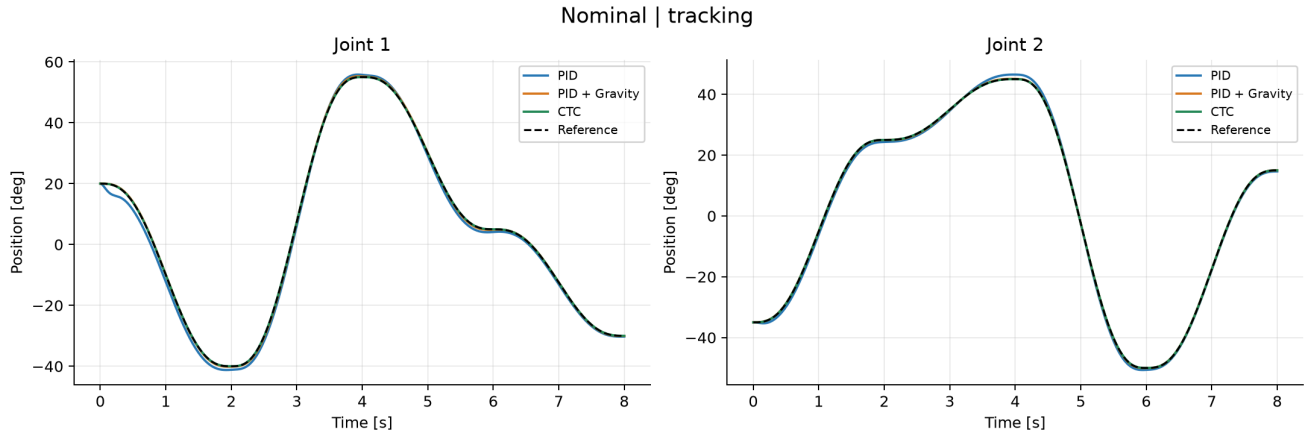


Figure 1: Nominal joint positions and reference trajectories; all controllers share the reference and initial state.

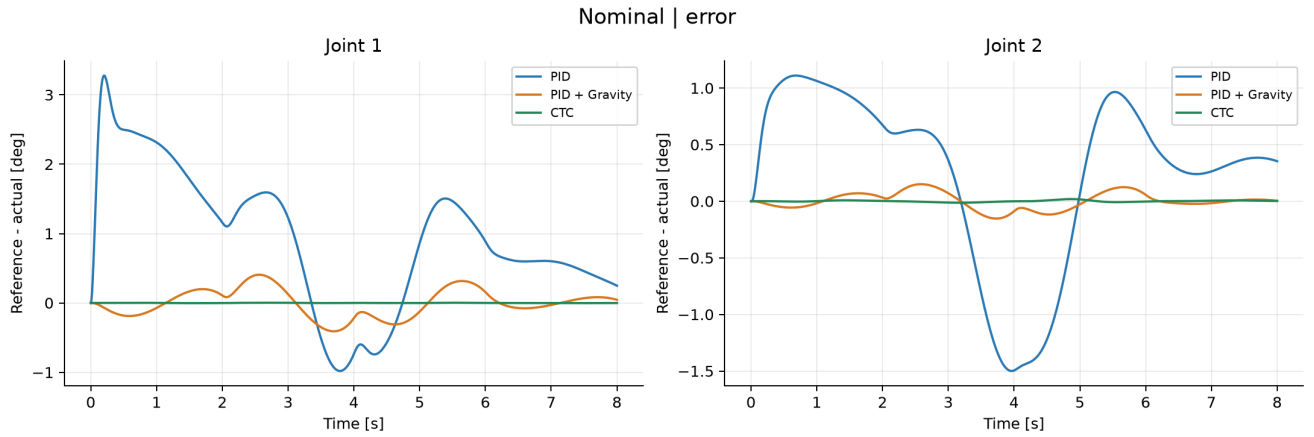


Figure 2: Nominal joint errors, retaining initial transients and every sampling interval.

6.2 Nominal Control Input and Combined Metrics

Figure 3 shows actual motor torques and shared limits; each controller has saturation fraction 0.000 / 0.000 / 0.000%. Figure 4 presents accuracy, effort, and mechanical work together. Small error-metric bars retain numerical labels rather than being described as exactly zero.

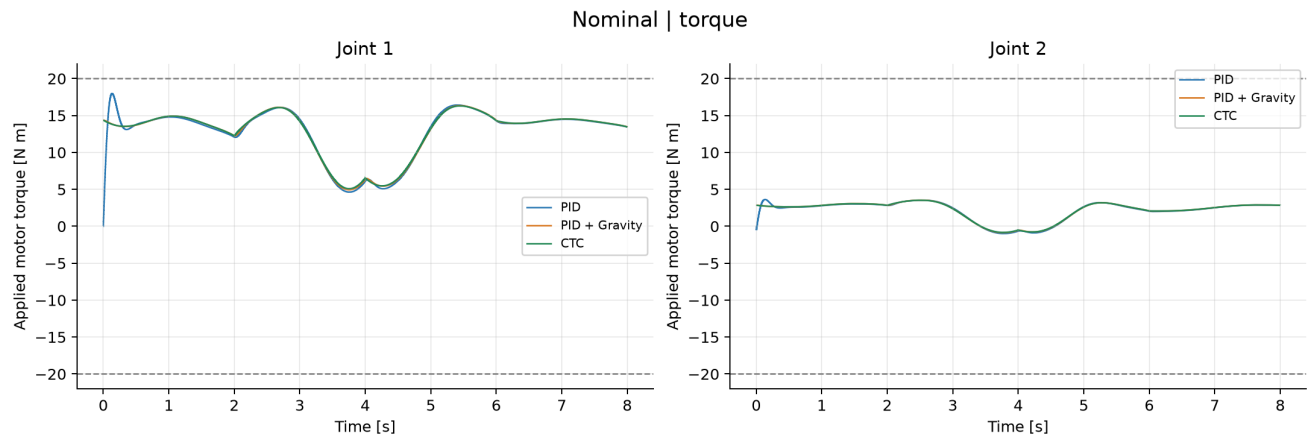


Figure 3: Actual nominal motor torque; dashed lines indicate shared actuator limits.

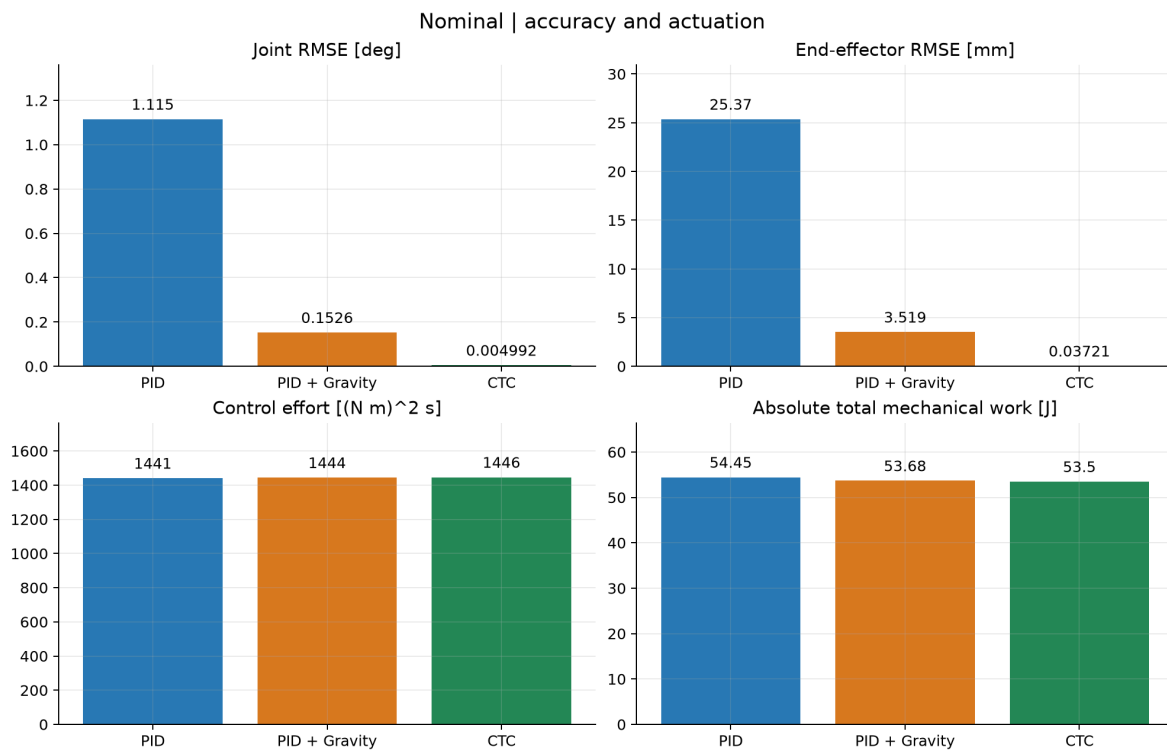


Figure 4: Nominal accuracy, control effort, and absolute total mechanical work, each in its own units.

6.3 Unknown Endpoint Payload

Figure 5 includes the full mass grid. At 0.25 kg, overall joint RMSE values are 1.381 / 0.339 / 3.967 degrees. At the maximum 1.0 kg payload, they reach 32.491 / 32.553 / 40.108 degrees, with saturation fractions 95.513 / 92.375 / 64.312%. These degraded runs remain in the analysis.

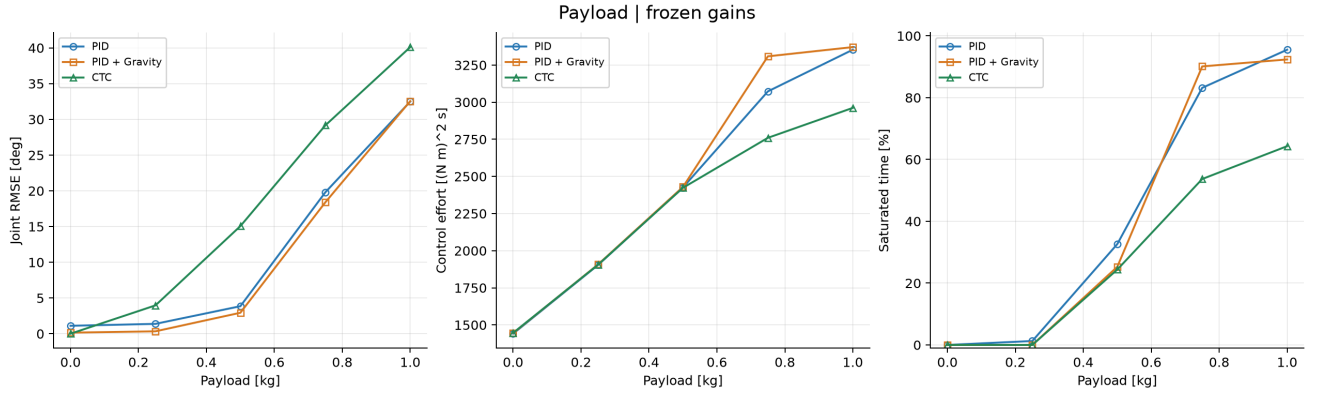


Figure 5: Payload effects on overall joint error, control effort, and saturation fraction.

Table 6: All payload joint RMSE values in degrees; controller order is fixed.

Payload (kg)	PID	PID+G	CTC
0.0	1.115037	0.152634	0.004992
0.25	1.380694	0.338827	3.967051
0.5	3.851948	2.936610	15.061124
0.75	19.789138	18.388365	29.155920
1.0	32.490911	32.552774	40.108152

At maximum payload, RMSE ratios relative to each controller's own nominal result are 29.139 / 213.273 / 8034.353. CTC's very small nominal denominator magnifies relative deterioration; absolute errors are therefore also reported. Joint and end-effector errors need not yield the same ranking, because large angular deviations change the geometric correspondence.

6.4 Controller-Model Parameter Error

In Figure 6, the plant is unchanged and both selected controller parameters are multiplied by $1 + \delta$. PID uses no internal dynamics model, so its RMSE remains 1.115037 degrees throughout the grid. CTC RMSE at -30% and +30% error is 2.828458 and 1.566751 degrees, respectively, versus 0.004992 degrees at zero error. Sensitivity is asymmetric.

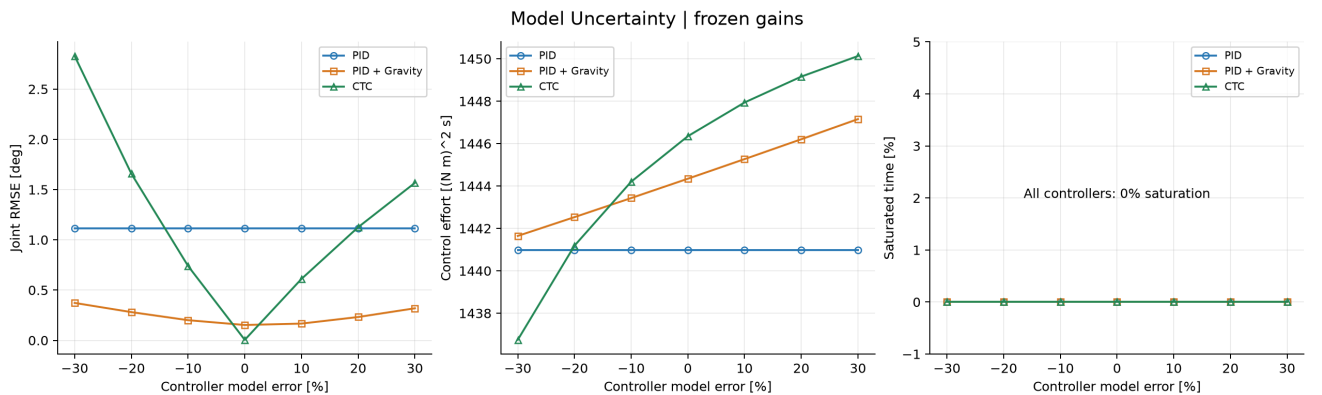


Figure 6: Shared relative error in controller-model mass and inertia; no controller saturates in this experiment.

6.5 External Endpoint Force

The base-frame force is $[10.0, 0.0]$ N during $[4.5, 4.7]$ s. The left panel of Figure 7 shows absolute reference error; the right shows additional deviation from the same controller's nominal trajectory. Their difference illustrates how baseline tracking error affects recovery time.

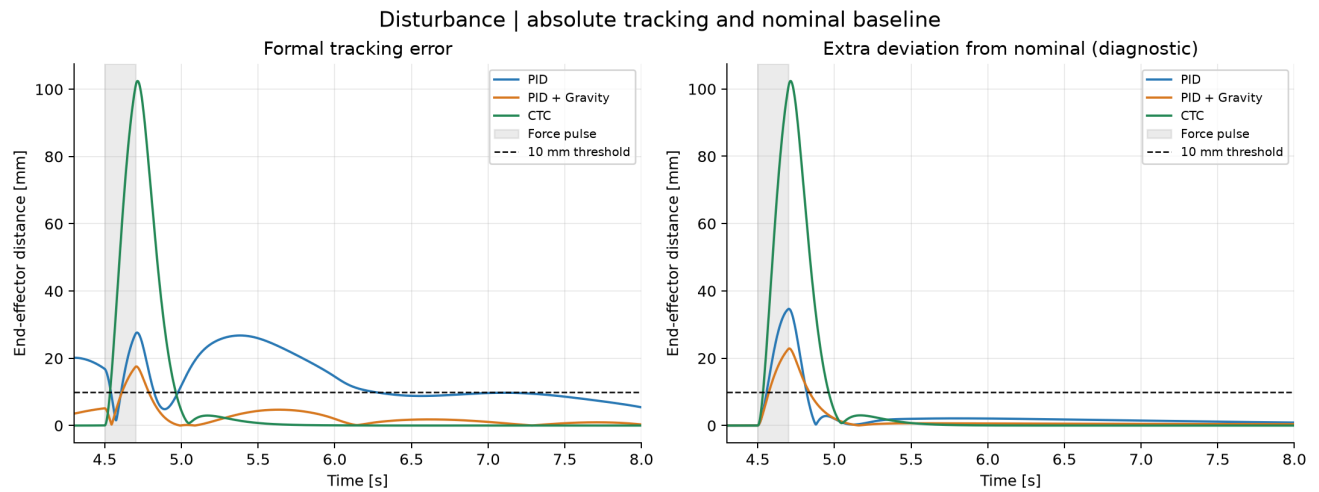


Figure 7: Force response: formal tracking error on the left and diagnostic nominal-relative deviation on the right.

Table 7: Disturbance recovery and peaks. Full-run and post-disturbance peaks are reported separately.

Controller	Recovery (s)	Full-run peak (mm)	Post-disturbance peak (mm)	Saturation %
PID	1.555	54.107	27.644	0.000
PID+G	0.087	17.564	17.564	0.000
CTC	0.263	102.448	102.448	0.000

All three controllers recover within the observation window, with formal delays of 1.555 / 0.087 / 0.263 s. PID with gravity compensation is fastest, followed by CTC and PID. Post-disturbance endpoint peaks are 27.644 / 17.564 / 102.448 mm. Fast recovery and small peak error must therefore be reported separately.

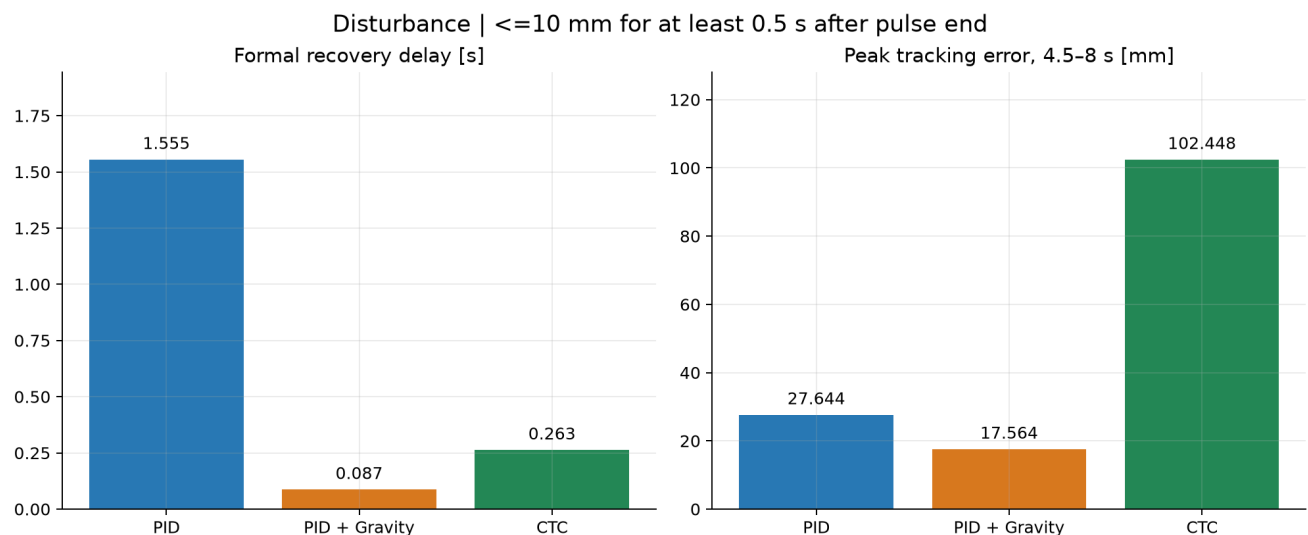


Figure 8: Formal recovery delay from pulse removal and peak error between disturbance onset and trajectory completion.

7 Discussion

7.1 Responses to the Charter Predictions

The nominal-accuracy prediction is supported by these data. CTC has the smallest overall joint RMSE with an accurate model. However, all controllers were tuned separately, so the full benefit cannot be attributed to adding a single compensation term. Structure and tuning jointly determine the observed differences.

The unknown-payload prediction requires a specified metric. CTC shows the strongest increase in joint RMSE relative to its own nominal baseline and larger joint RMSE at the tested nonzero payloads. End-effector rankings can differ because of geometric mapping and large angular deviations; CTC is not necessarily worst on every payload metric. Larger payloads also cause saturation, so mismatch and actuator capacity cannot be isolated in these runs.

The model-error prediction receives conditional support. CTC is best near zero error; its advantage decreases in either direction and it may underperform model-independent PID at larger mismatch. Positive and negative errors are asymmetric and should not be fitted by a single line in absolute error. Gravity-compensated PID degrades less across this grid, without guaranteeing the same behavior for other trajectories or parameter errors.

The recovery prediction is supported by the formal metric. Gravity-compensated PID meets the threshold first. PID's longer formal recovery includes baseline tracking error, which the diagnostic panel helps distinguish. External force still drives CTC error; high nominal accuracy does not guarantee smaller disturbance peaks or faster recovery.

7.2 Interpretation Boundaries

A best-controller claim must specify operating conditions and metrics. Nominal accuracy, parameter sensitivity, actuator saturation, disturbance peaks, and recovery time are different objectives. Squared-torque effort is a tuning penalty, not electrical energy. Mechanical work also excludes motor and drive losses and regenerative-braking efficiency. These data therefore do not support energy-saving claims.

8 Limitations

The study covers one fixed robot, training and held-out trajectories, finite payload and error grids, and one force direction and pulse interval. It excludes friction, backlash, delay, flexibility, and sensor noise; controllers receive true joint states. The finite candidate budget does not establish global optimality, and parameterizations differ. Fixed-step integration and sampled holds introduce discretization effects. All findings are simulation results, without hardware validation or universal guarantees for untested conditions or closed-loop stability.

Future work could add stochastic repeats, noise, multiple force directions, other trajectories, and matched-gain ablations. These extensions are not used to retune the current controllers. Failed and degraded conditions remain visible, without selection based on the expected conclusion.

9 Conclusion

With frozen gains and shared test conditions, CTC achieves the highest nominal tracking accuracy. Unknown payload, model error, and force disturbances change the relationship among controllers. Gravity-compensated PID recovers fastest under the formal disturbance criterion. These data support no single winner across every condition and metric. Traceable sources, automatic numerical assembly, fixed figures, and HTML/PDF outputs from one source make the results reviewable.

10 Reproducibility and Sources

`stage22/build_report.py` assembles the report from fixed Stage 21 outputs and original configurations, then Quarto renders HTML and Typst PDF from the same source. HTML embeds resources and uses MathML; the English PDF uses Arial. The rendering workflow follows the official Quarto Typst documentation and needs no separate LaTeX installation. Raw results are not modified, and failed builds are not marked as accepted.

From the project root, with the project environment activated:

```
python stage22/build_report.py --output stage22/rebuilt_report
```

Existing output directories are protected. Use `--output` to select a new directory. The package includes the full CSV, eight source figures, generated Quarto source, HTML, PDF, and a build manifest with input, program, and output hashes. Raw NPZ/YAML paths in the CSV are relative to this project root. Tables display rounded values; the CSV retains full precision.

Table 8: Fixed run directories used by this report.

Scenario	Run directory under results/
nominal	20260907T024619_815442Z
payload	20260911T014147_244577Z
model_uncertainty	20260911T025328_299278Z
disturbance	20260913T114926_385587Z

References and Project Evidence

- Modern Robotics, Chapter 11.4: computed torque and dynamic model compensation; linked in the introduction.
- Official Quarto Typst documentation: PDF rendering and font configuration; linked above.
- [RESEARCH_CHARTER.md](#): predefined research questions and four predictions.
- Frozen [results/tuning_search.json](#): all candidates, objective, seed, and final selections.
- Stage 21 summary, figures, and manifest: experimental results and their source provenance.

11 Full Results

These tables retain all 42 runs. Joint RMSE is in degrees; endpoint RMSE and full-run peak are in mm; saturation is in %. The accompanying [summary.csv](#) retains full fields for effort, absolute mechanical work, recovery status, normalized results, and source paths.

11.1 nominal

Parameter	Controller	Joint RMSE	EE RMSE	Full-run peak	Saturation %
—	PID	1.115037	25.366	54.107	0.000
—	PID+G	0.152634	3.519	7.221	0.000
—	CTC	0.004992	0.037	0.101	0.000

11.2 payload

Parameter	Controller	Joint RMSE	EE RMSE	Full-run peak	Saturation %
0.0	PID	1.115037	25.366	54.107	0.000
0.0	PID+G	0.152634	3.519	7.221	0.000
0.0	CTC	0.004992	0.037	0.101	0.000
0.25	PID	1.380694	31.104	67.972	1.275
0.25	PID+G	0.338827	7.785	15.167	0.000
0.25	CTC	3.967051	28.539	41.753	0.000
0.5	PID	3.851948	81.910	249.523	32.625
0.5	PID+G	2.936610	60.548	213.262	25.275
0.5	CTC	15.061124	68.595	163.053	24.375
0.75	PID	19.789138	401.421	780.065	83.188
0.75	PID+G	18.388365	371.861	825.579	90.112
0.75	CTC	29.155920	153.646	394.210	53.688
1.0	PID	32.490911	642.289	1246.316	95.513
1.0	PID+G	32.552774	635.600	1229.850	92.375
1.0	CTC	40.108152	264.565	586.778	64.312

11.3 model_uncertainty

Parameter	Controller	Joint RMSE	EE RMSE	Full-run peak	Saturation %
-0.3	PID	1.115037	25.366	54.107	0.000
-0.3	PID+G	0.372387	8.519	15.129	0.000
-0.3	CTC	2.828458	38.008	52.026	0.000
-0.2	PID	1.115037	25.366	54.107	0.000
-0.2	PID+G	0.280691	6.442	12.492	0.000
-0.2	CTC	1.657619	22.424	30.459	0.000
-0.1	PID	1.115037	25.366	54.107	0.000
-0.1	PID+G	0.200772	4.629	9.857	0.000
-0.1	CTC	0.739656	10.056	13.572	0.000
0.0	PID	1.115037	25.366	54.107	0.000
0.0	PID+G	0.152634	3.519	7.221	0.000

Parameter	Controller	Joint RMSE	EE RMSE	Full-run peak	Saturation %
0.0	CTC	0.004992	0.037	0.101	0.000
0.1	PID	1.115037	25.366	54.107	0.000
0.1	PID+G	0.166760	3.797	7.584	0.000
0.1	CTC	0.611066	8.335	11.144	0.000
0.2	PID	1.115037	25.366	54.107	0.000
0.2	PID+G	0.232168	5.248	10.190	0.000
0.2	CTC	1.125612	15.360	20.461	0.000
0.3	PID	1.115037	25.366	54.107	0.000
0.3	PID+G	0.318798	7.198	13.659	0.000
0.3	CTC	1.566751	21.361	28.369	0.000

11.4 disturbance

Parameter	Controller	Joint RMSE	EE RMSE	Full-run peak	Saturation %
—	PID	1.103720	25.197	54.107	0.000
—	PID+G	0.176402	3.997	17.564	0.000
—	CTC	2.237485	15.139	102.448	0.000